

Health Facility Assessment (HFA) for small and sick newborns

Neonatal Unit Infrastructure Module

Version: February 2, 2023

Date (yyyy/mm/dd): / /

Data Collector 1 (first and last name):

Data Collector 2 (first and last name):

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Facility Name:

Facility Code:

Module Overview for Neonatal Unit Infrastructure

Section 1. Neonatal Unit Infrastructure (direct questions to nurse in charge)

Section 2. Neonatal Unit Infection Prevention and Control (direct questions to nurse in charge)

Section 3. Neonatal Unit Electricity (direct questions to nurse in charge and technician)

Section 4. Neonatal Unit Layout

Section 5. Admission and Referral Criteria (direct questions to nurse in charge)

Appendix A: Neonatal Unit Layout Guidance

Facility Name:

Facility Code:

Section 1. Neonatal Unit Infrastructure

Instructions: Copy the Facility Code onto each page before the team begins to collect data. Please have all participants fill in their names and positions below, before completing the following forms in *Section 1. Neonatal Unit Infrastructure*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 1. Neonatal Unit Infrastructure

Instructions: Direct these questions to the nurse in charge of the neonatal unit. For some questions, you will need to ask and observe the staff or specific items/spaces at the facility. An item is not functional if it is in need of repair. Please do not leave any variable blank. If a response is unknown, write “unknown” or select “unknown” from the multiple-choice answers.

Prompt: “I'd like to ask you a few questions and observe the neonatal unit's infrastructure.”

No.	Item	Response	Skip to
Neonatal Unit Capacity			
1.01	Does the facility provide neonatal care 24 hours a day, 7 days a week?	☐ Yes ☐ No	—
1.02	What is the total capacity of your neonatal unit if you have one baby per bed, including cots, radiant warmers, incubators, etc.?	Number:	—
1.03	How many baby cots do you have in your neonatal unit? (Count all mobile and stationary baby cots)	Number:	—
1.04	Of all baby cots, how many are mobile baby cots with castors?	Number:	If “0”, skip to 1.06
1.05	Of the mobile baby cots, how many are functional for a baby to use?	Number:	—
1.06	Of all baby cots, how many are stationary baby cots without castors?	Number:	If “0”, skip to 1.08
1.07	Of the stationary baby cots, how many are functional for a baby to use?	Number:	—
1.08	Is there a mattress for every baby cot?	☐ Yes ☐ No	If “Yes”, skip to 1.10
1.09	If no, how many mattresses are there? (Count all mattresses)	Number:	—
1.10	Do all the baby cots have insecticide treated bed nets?	☐ Yes ☐ No	—
1.11	How close to each other are the closest two cots? (Measure the space between the sides/walls of the two closest cots in centimeters)	cm	—
1.12	Do you ever have more than one baby per cot ? If yes, how often? (Select one) - Frequently: More than 2 times every week - Sometimes: Less than 1 time every week - Rarely: Less than 1 time every month	☐ Frequently ☐ Sometimes ☐ Rarely ☐ Never ☐ N/A (no cots)	—
1.13	Do you ever have more than one baby per radiant warmer ? If yes, how often? (Select one) - Frequently: More than 2 times every week - Sometimes: Less than 1 time every week - Rarely: Less than 1 time every month	☐ Frequently ☐ Sometimes ☐ Rarely ☐ Never ☐ N/A (no radiant warmer)	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
1.14	Do you ever have more than one baby per incubator ? If yes, how often? <i>(Select one)</i> <i>Frequently: More than 2 times every week</i> <i>Sometimes: Less than 1 time every week</i> <i>Rarely: Less than 1 time every month</i>	☐ Frequently ☐ Sometimes ☐ Rarely ☐ Never ☐ N/A (no incubator)	—
1.15	Do you ever have more than one baby per phototherapy light ? If yes, how often? <i>(Select one)</i> <i>Frequently: More than 2 times every week</i> <i>Sometimes: Less than 1 time every week</i> <i>Rarely: Less than 1 time every month</i>	☐ Frequently ☐ Sometimes ☐ Rarely ☐ Never ☐ N/A (no phototherapy light)	—
1.16	How many babies are currently in the neonatal unit? <i>(Count the number of babies currently in the neonatal unit. Do not count babies in the Kangaroo Mother Care (KMC) unit.)</i>	Number:	—
1.17	How many babies are currently under phototherapy lights in the neonatal unit? <i>(Count the number of babies currently under phototherapy lights in the unit)</i>	Number:	—
1.18	How many babies are currently under radiant warmers in the neonatal unit? <i>(Count the number of babies currently under radiant warmers in the unit)</i>	Number:	—
1.19	How many babies are currently in incubators on the neonatal unit? <i>(Count the number of babies currently in incubators in the unit)</i>	Number:	—
1.20	Did you ever refuse admission or refer a neonate to another facility because of inadequate number of beds in your neonatal unit in the last year?	☐ Yes ☐ No	—
1.21	Do you ever admit neonates in the paediatric ward?	☐ Yes ☐ No	—
Transfers			
1.22	How do you usually transfer the neonates from the labour ward to the neonatal unit? <i>(Select all that apply)</i>	☐ Carried by hand ☐ Carried in plastic wrap ☐ Carried by KMC ☐ Carried in an incubator/radiant warmer ☐ Carried in a cot ☐ Other <i>(specify below)</i>	—
1.23	Is the neonatal unit in the same building as the labour ward? <i>(Select NO if the units are located in different buildings, even if they are connected by a covered walkway with no walls. If you have to walk outside, the units are located in different buildings.)</i>	☐ Yes ☐ No ☐ N/A (no labour ward)	— — —

Facility Name:

Facility Code:

No.	Item	Response	Skip to
1.24	Is the neonatal unit in the same building as the maternity theatre? <i>(Select NO if the units are located in different buildings, even if they are connected by a covered walkway with no walls. If you have to walk outside, the units are located in different buildings.)</i>	☐ Yes ☐ No ☐ N/A (no maternity theatre)	—
Dedicated Space Observations			
1.25	Which of the following areas do you have in the neonatal unit? <i>(Read all answer options aloud. Select all that apply.)</i>	☐ High risk/acute area ☐ Low risk/stable area ☐ Area for those admitted from outside the facility ☐ Isolation room ☐ None of the above ☐ Other <i>(specify below)</i>	—
			—
1.26	Is there a dedicated area for examination and triage when a newborn first arrives in the unit?	☐ Yes ☐ No ☐ Yes, but not observed	—
Neonatal Unit Temperature			
1.27	Which, if any, functional heating arrangements are there for the neonatal unit? <i>(Select all that apply)</i> <i>(This does not include a radiant warmer or heating for neonates. Central heating is a system of heating buildings by warming air or water at one place and then sending it to different rooms through pipes or vents. When combined with other systems to control the building temperature (for example, cooling) the whole system can be an HVAC system.)</i>	☐ None ☐ Central heating ☐ Electric space heater ☐ Gas heater ☐ Other <i>(specify below)</i>	—
			—
1.28	How many electric space heaters are available ?	Number:	If "0" skip to 1.31
1.29	How many electric space heaters are functional ?	Number:	—
1.30	<i>For observation only:</i> Are space heaters placed appropriately? <i>(Appropriate placement means there are no cables across the floor, the heaters are at a safe distance from babies, and not touching items that could catch fire.)</i>	☐ Yes ☐ No	—
1.31	Is there a functional room thermometer within the neonatal unit?	☐ Yes ☐ No ☐ Yes, but not observed	—
1.32	How many rooms within the neonatal unit have functional room thermometers?	Number:	If "0", skip to 1.35
1.33		Room 1: °C	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
	What is the ambient room temperature in the room(s) with functional thermometers? <i>(Enter a number for the ambient room temperature in degrees Celsius to one decimal place. If more than three rooms, add additional readings in comments below).</i>	Room 2: °C Room 3: °C	
1.34	For observation only: What time was the ambient room temperature observed?		—
1.35	For observation only: Observe if there is a draught coming from any of the windows (e.g., broken or open window).	☐ Yes ☐ No	If "No", 1.37
1.36	Comments on draught from windows:		
Lighting			
1.37	Is there adequate lighting to ensure good illumination during the day? <i>(There should be enough lighting for staff to easily carry out their work and complete any documentation at any time of day)</i>	☐ Yes ☐ No	—
1.38	Is there adequate lighting to ensure good illumination during the night? <i>(There should be enough lighting for staff to easily carry out their work and complete any documentation at any time of night)</i>	☐ Yes ☐ No	—
Medical Gases Systems			
1.39	Are there any oxygen wall outlets/ports in the neonatal unit?	☐ Yes ☐ No	If "No", 1.43
1.40	How many oxygen wall outlets/ports are there in the neonatal unit?	Number:	—
1.41	Is there walled/piped blended oxygen in the neonatal unit? <i>(Blended oxygen is a mixture of medical grade oxygen and room air. It is used to regulate the concentration of oxygen delivered.)</i>	☐ Yes ☐ No ☐ Yes, but not observed	—
1.42	Is there walled/piped medical air in the neonatal unit? <i>(Medical air is a clean supply of compressed or ambient air, which is free of contaminants and impurities.)</i>	☐ Yes ☐ No ☐ Yes, but not observed	
Nurses Station			
1.43	Is there a nurse's station or staff work area within the neonatal unit?	☐ Yes ☐ No	If "No", 1.46
1.44	Is there a place for staff to sit (e.g., chair, bench, <i>not</i> on the floor)?	☐ Yes ☐ No	—
1.45	Can the nurses see every baby from the nurses' station?	☐ Yes ☐ No	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
Water			
1.46	Is there a time of the year when the neonatal unit routinely has a shortage or lack of water?	☐ Yes ☐ No	—
1.47	Is there a reliable backup source of water? <i>(Reliable is defined as providing the facility with water supply to meet all its needs while the main water source is not functioning)</i>	☐ Yes ☐ No	—
Fire Prevention			
1.48	Is there a fire extinguisher?	☐ Yes ☐ No	If "No", 1.51
1.49	Has the fire extinguisher ever been formally inspected for functionality?	☐ Yes ☐ No ☐ Unknown	—
1.50	When was it last inspected? (yyyy/mm/dd) <i>(If date is unknown, enter 1999/01/01)</i>	____/____/____	—
1.51	Does the facility conduct fire drills at least once per year?	☐ Yes ☐ No ☐ Unknown	—
1.52	Does the facility have a neonatal evacuation plan in case of a fire?	☐ Yes ☐ No ☐ Unknown	—
Emergency Drills			
1.53	Has there been an emergency care drill on the neonatal unit in the last 12 months?	☐ Yes ☐ No ☐ Unknown	—

Facility Name:

Facility Code:

Comments

Facility Name:

Facility Code:

Section 2. Neonatal Unit Infection Prevention and Control

Instructions: Please have all participants fill in their names and positions below, before completing the following forms in *Section 2. Neonatal Unit Infection Prevention and Control*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 2. Neonatal Unit Infection Prevention and Control

Instructions: Direct these questions to the nurse in charge. For some questions, you will need to observe the staff or specific items/spaces at the facility. If a guideline is available, continue to ask if it is easily accessible. All guidelines must be seen in order to select "Yes" to being available and to being easily accessible. In order to select "Yes" to easily accessible the nurse must know where the guideline is stored and be able to retrieve it quickly. Please do not leave any variable blank. If a response is unknown, write "unknown" or select "unknown" from the multiple-choice answers.

Prompt: "Next I'll ask you a few questions about the neonatal unit's infection prevention."

2A. Guidelines

No.	Does the facility have the following?	Available and easily accessible	Available but not easily accessible	Not available
2.01	Infection Prevention Protocols			
2.01a	Prevention of neonatal infections protocol	☐	☐	☐
2.01b	Handwashing wall chart	☐	☐	☐
2.01c	Cleaning the neonatal unit guidelines	☐	☐	☐
2.01d	Cleaning medical equipment guidelines	☐	☐	☐
2.02	Infection Control Protocols			
2.02a	Infection control guidelines	☐	☐	☐
2.02b	Early diagnosis and management of neonatal infections protocol	☐	☐	☐
2.02c	Drug doses, dilutions and preparations for the neonatal unit guidelines	☐	☐	☐
2.02d	Safe and rational use of antibiotics and other medications based on weight and age guidelines (e.g. antibiotic stewardship)	☐	☐	☐
2.02e	Prescription chart	☐	☐	☐
2.02f	Care of small and sick newborns in outbreak guidelines	☐	☐	☐
2.02g	Accidental needle prick protocol	☐	☐	☐

No.	Item	Response	Skip to
	Surveillance		
2.03	Is surveillance conducted for device-associated infections (for example, catheter-associated urinary tract infections, central line-associated bloodstream infections)?	☐ Yes ☐ No	—
2.04	Is surveillance conducted for hospital-acquired infection outbreaks for neonates?	☐ Yes ☐ No	—

No.	Item	Response	Skip to
	Hand hygiene policies		
2.05	Is there a policy for staff to wash their hands with soap before or immediately upon entry to the neonatal unit?	☐ Yes ☐ No	—
2.06	Is there a policy for staff to use hand sanitiser before or immediately upon entry to the neonatal unit?	☐ Yes ☐ No	—
2.07	Is there a policy for visitors to wash their hands with soap and/or use hand sanitiser before or immediately upon entry to the neonatal unit?	☐ Yes ☐ No	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
Hand hygiene before or immediately at the entrance to the neonatal unit			
2.08	Are hand hygiene options available before or immediately at the entrance to the neonatal unit?	☐ Yes ☐ No	If "No", 2.12
2.09	What hand hygiene options are available before or immediately at the entrance to the neonatal unit? (Select all that apply)	☐ Sink with soap ☐ Sink without soap ☐ Sink with hand sanitiser ☐ Free-standing hand sanitiser or alcohol-based rub ☐ Bucket ☐ Other (specify below)	—
			—
2.10	How are water taps before or immediately at the entrance operated? (Select all that apply)	☐ Hand ☐ Elbow ☐ Foot ☐ Knee ☐ N/A (no taps)	—
2.11	What, if any, options for hand drying are available before or immediately at the entrance to the neonatal unit? (Select all that apply)	☐ None ☐ Single use hand drying towels ☐ Cloth towel ☐ Electric hand dryer ☐ Other (specify below)	—
			—
Hand hygiene before entry to and in each room in the neonatal unit			
2.12	Are hand hygiene options available before entry to and in each room in the neonatal unit?	☐ All rooms ☐ Some rooms ☐ No rooms	—
All Sinks			
2.13	For observation only: Are all of the sinks functioning? (All sinks should have running water, not be clogged up, not have leaks)?	☐ All sinks ☐ Some sinks ☐ No sinks	—
2.14	Have liquid soaps been diluted with water? (Observe all soaps in the neonatal unit)	☐ All ☐ Some ☐ None ☐ N/A (no soap)	—
No.	Item	Response	Skip to
	Toilets/Latrines		

Facility Name:

Facility Code:

No.	Item	Response	Skip to
2.15	Are there toilets for staff or caretaker/guardian use? (Select all that apply)	☐ No toilets ☐ Yes, separate toilet for staff ☐ Yes, separate toilet for caretaker/guardian ☐ Yes, shared toilet for staff and caretaker/guardian	—
Separate staff toilets			
2.16	Can staff access the toilet at the time of the visit?	☐ Yes ☐ No	—
2.17	Can staff access the toilet 24 hours per day 7 days per week?	☐ Yes ☐ No	—
2.18	Is the toilet for staff use visibly clean and functional? (Functioning means that it does not require any maintenance for appropriate use)	☐ Yes ☐ No ☐ Yes, but not observed	—
2.19	What hand hygiene options are available in the staff toilet? (Select all that apply)	☐ None ☐ Sink with soap ☐ Sink without soap ☐ Sink with hand sanitiser ☐ Free-standing hand sanitiser or alcohol-based rub ☐ Bucket ☐ Other (specify below)	—
			—
2.20	What, if any, options for hand drying are available in the staff toilet? (Select all that apply)	☐ None ☐ Single use hand drying towels ☐ Cloth towel ☐ Electric hand dryer ☐ Other (specify below)	—
			—
2.21	Location of nearest toilet for staff use from the neonatal unit: (Select one)	☐ Neonatal unit ☐ Postnatal unit ☐ KMC unit ☐ Elsewhere in the facility ☐ Not on premises	—
2.22	What type of toilet or latrine is it? (Select one)	☐ Flush or pour flush toilet ☐ Ventilated improved pit latrine ☐ Pit latrine with slab ☐ Pit latrine without slab/open pit ☐ Composting toilet ☐ Unknown ☐ Other (specify below)	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
			—
2.23	How often are toilets for staff use cleaned? (Select one)	☐ Daily ☐ Weekly ☐ Irregularly ☐ Never ☐ Other (specify below)	—
			—
2.24	Is there a rota of times they are cleaned?	☐ Yes ☐ No ☐ Yes, but not observed	—
2.25	Is there a dedicated person whose job it is to clean the staff toilets?	☐ Yes ☐ No	—
Separate caretaker/guardian toilet			
2.26	Can caretakers/guardians access the toilet at the time of the visit?	☐ Yes ☐ No	—
2.27	Which family members are permitted to use the toilet? (Select all that apply)	☐ Mother ☐ Father ☐ Grandmother ☐ Caretaker/ Guardian ☐ Other (specify below)	—
			—
2.28	Can caretakers/guardians access the toilet 24 hours per day 7 days per week?	☐ Yes ☐ No	—
2.29	Is the toilet for caretaker/guardian use visibly clean and functional? (Functioning means that it does not require any maintenance for appropriate use)	☐ Yes ☐ No ☐ Yes, but not observed	—
2.30	What hand hygiene options are available in the caretaker/guardian toilet? (Select all that apply)	☐ None ☐ Sink with soap ☐ Sink without soap ☐ Sink with hand sanitiser ☐ Free-standing hand sanitiser or alcohol-based rub ☐ Bucket ☐ Other (specify below)	—
			—
2.31	What, if any, options for hand drying are available in the caretaker/guardian toilet? (Select all that apply)	☐ None ☐ Single use hand drying towels ☐ Cloth towel ☐ Electric hand dryer ☐ Other (specify below)	—
			—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
2.32	Location of nearest toilet for caretaker/guardian use from the neonatal unit: (Select one)	☐ Neonatal unit ☐ Postnatal unit ☐ KMC unit ☐ Elsewhere in the facility ☐ Not on premises	—
2.33	What type of toilet or latrine is it? (Select one)	☐ Flush or pour flush toilet ☐ Ventilated improved pit latrine ☐ Pit latrine with slab ☐ Pit latrine without slab/open pit ☐ Composting toilet ☐ Unknown ☐ Other (specify below)	—
			—
2.34	How often are toilets for caretaker/guardian use cleaned? (Select one)	☐ Daily ☐ Weekly ☐ Irregularly ☐ Never ☐ Other (specify below)	—
			—
2.35	Is there a rota of times they are cleaned?	☐ Yes ☐ No ☐ Yes, but not observed	—
2.36	Is there a dedicated person whose job it is to clean the caretaker/guardian toilets?	☐ Yes ☐ No	—
Shared staff and caretaker/guardian toilet			
2.37	Can staff and caretakers/guardians access the shared toilet at the time of the visit?	☐ Yes ☐ No	—
2.38	Which family members are permitted to use the shared toilet? (Select all that apply)	☐ Mother ☐ Father ☐ Grandmother ☐ Caretaker/Guardian ☐ Other (specify below)	—
			—
2.39	Can staff and caretakers/guardians access the shared toilet 24 hours per day 7 days per week?	☐ Yes ☐ No	—
2.40	Is the shared toilet for staff and caretaker/guardian use visibly clean and functional? (Functioning means that it does not require any maintenance for appropriate use)	☐ Yes ☐ No ☐ Yes, but not observed	—
2.41	What hand hygiene options are available in the shared toilet? (Select all that apply)	☐ None ☐ Sink with soap ☐ Sink without soap ☐ Sink with hand sanitiser	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
		☐ Free-standing hand sanitiser or alcohol-based rub ☐ Bucket ☐ Other (<i>specify below</i>)	
			—
2.42	What, if any, options for hand drying are available in the staff and caretaker/guardian toilet? (<i>Select all that apply</i>)	☐ None ☐ Single use hand drying towels ☐ Cloth towel ☐ Electric hand dryer ☐ Other (<i>specify below</i>)	—
			—
2.43	Location of nearest shared toilet for staff and caretaker/guardian use from the neonatal unit: (<i>Select one</i>)	☐ Neonatal unit ☐ Postnatal unit ☐ KMC unit ☐ Elsewhere in the facility ☐ Not on premises	—
2.44	What type of toilet or latrine is it? (<i>Select one</i>)	☐ Flush or pour flush toilet ☐ Ventilated improved pit latrine ☐ Pit latrine with slab ☐ Pit latrine without slab/open pit ☐ Composting toilet ☐ Unknown ☐ Other (<i>specify below</i>)	—
			—
2.45	How often are shared toilets cleaned? (<i>Select one</i>)	☐ Daily ☐ Weekly ☐ Irregularly ☐ Never ☐ Other (<i>specify below</i>)	—
			—
2.46	Is there a rota of times they are cleaned?	☐ Yes ☐ No ☐ Yes, but not observed	—
2.47	Is there a dedicated person whose job it is to clean the shared toilets?	☐ Yes ☐ No	—

No.	Item	Response	Skip to
	Sterilisation and Equipment Disinfection		
2.48	Is there an area for cleaning and disinfecting supplies and equipment on the neonatal unit?	☐ Yes ☐ No ☐ Yes, but not observed	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
2.49	Are disinfection buckets available on the day of the visit?	☐ Yes ☐ No ☐ Yes, but not observed	—
2.50	What areas are there for cleaning and disinfecting supplies and equipment on the neonatal unit? (Select all that apply)	☐ None ☐ Separate sluice or soiled utility room ☐ Separate sink used exclusively for supplies and equipment ☐ Sink, but shared for hand washing ☐ Disinfection bucket ☐ Other (specify below)	—
2.51	Is there a space for sterilisation? (For example, space for dry heat, steam, electric boiler, steamer, non-electrics with cover for boiling and steaming & drum and stand)	☐ Yes ☐ No ☐ Yes, but not observed	—
2.52	Is there a storage space on the neonatal unit for supplies, including linen, clinical supplies, and general equipment)?	☐ Yes ☐ No	—
2.53	For observation only: Is the storage space clean and tidy, and organised? (Items should be neatly stacked, easily visible, and accessible, the space should not be over filled, and it should be free of spills)	☐ Yes ☐ No	—

No.	Item	Response	Skip to
Ventilation			
2.54	Is there a functioning air filtration system to prevent risk of airborne infection? (An air filtration system can include any freestanding equipment or building system that filters the air in the neonatal unit)	☐ Yes ☐ No	—

No.	Item	Response	Skip to
Waste Management			
2.55	Is there a functioning incinerator (e.g., burning incinerator, non-burning incinerator, off-site incineration)? (May be shared with other units)	☐ Yes ☐ No ☐ Yes, but not observed	—
2.56	What types of separate covered waste bins are available on the neonatal unit? (Select all that apply)	☐ Puncture-proofs sharps container ☐ Contaminated waste (red) ☐ Discarded medicines (black) ☐ Anatomical/human waste (yellow) ☐ Glass (blue) ☐ General trash bin ☐ Other (specify below)	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
			—
2.57	For observation only: Are waste bins used appropriately on the day of the visit? (Appropriate use of waste bins includes the covering of bins with a lid, correct designation of waste, and bins not overflowing)	☐ All ☐ Some ☐ None	—
2.58	Is trash collected at least once per day from the neonatal unit?	☐ Yes ☐ No	—
2.59	Are there designated staff for waste handling?	☐ Yes ☐ No	—
2.60	Is there storage space for soiled utility?	☐ Yes ☐ No	—
2.61	For observation only: Can you see any liquid spills or trash on the floor in the neonatal unit?	☐ Yes ☐ No	—
2.62	Do you ever see any vermin like mice or cockroaches, or animal feces on the neonatal unit?	☐ Yes ☐ No	—
2.63	Is there a staff rest area (e.g. tea room)?	☐ Yes ☐ No	—
2.64	Is food allowed in patient areas?	☐ Yes ☐ No	—

No.	Item	Response	Skip to
	Personal Items		
2.65	How many neonatal unit uniforms do you own?	Number:	—
2.66	Is there a place for staff to leave their personal items (e.g., lockers, cubbies, etc.)? (Select one)	☐ Yes ☐ No ☐ Yes, but not observed	—
2.67	Are staff mobile phones allowed in the neonatal unit? (Select one)	☐ Yes, any phones ☐ Yes, if disinfected before entry ☐ No ☐ No regulations for mobile phones	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
2.68	Are caretaker/guardian mobile phones allowed in the neonatal unit? (Select one)	☐ Yes, any phones ☐ Yes, if disinfected before entry ☐ No ☐ No regulations for mobile phones	—

Comments

Facility Name:

Facility Code:

Section 3. Neonatal Unit Electricity

Instructions: Please have all participants fill in their names and positions below, before completing the following forms in *Section 3. Neonatal Unit Electricity*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 3. Neonatal Unit Electricity

Instructions: Direct these questions to the nurse in charge. For some questions, you will need to observe the staff or specific items/spaces at the facility. Please do not leave any variable blank. If a response is unknown, write “unknown” or select “unknown” from the multiple-choice answers.

No.	Item	Response	Skip to
Fuel-operated Generator			
3.01	Does this facility have a fuel-operated generator?	☐ Yes ☐ No ☐ Yes, but not observed	If “No”, 3.07
3.02	Is the neonatal unit covered by this generator?	☐ Yes ☐ No	—
3.03	Does this generator exclusively cover the neonatal unit?	☐ Yes ☐ No ☐ Unknown	—
3.04	What neonatal unit lighting does the fuel-operated generator cover?	☐ All lighting ☐ Some lighting ☐ No lighting	—
3.05	What neonatal unit equipment does the fuel-operated generator cover?	☐ All equipment ☐ Some equipment ☐ No equipment	If All or No, 3.07
3.06	Specify what neonatal unit equipment is powered by the fuel-operated generator: (Select all that apply)	☐ Oxygen Concentrator ☐ Oxygen Flow Splitter ☐ CPAP ☐ Suction Pump ☐ Syringe Pump ☐ Phototherapy Light ☐ Radiant Warmer ☐ Incubator ☐ Other (specify below)	—
Battery Inverter			
3.07	Does this facility have a battery inverter?	☐ Yes ☐ No ☐ Yes, but not observed	If “No”, 3.13
3.08	Is the neonatal unit covered by the battery inverter?	☐ Yes ☐ No	—
3.09	Does the battery inverter exclusively cover the neonatal unit?	☐ Yes ☐ No ☐ Unknown	—
3.10	What neonatal unit lighting does the battery inverter cover?	☐ All lighting ☐ Some lighting ☐ No lighting	—
3.11	What neonatal unit equipment does the battery inverter cover?	☐ All equipment ☐ Some equipment ☐ No equipment	If All or No, 3.13
3.12	Specify what neonatal unit equipment is powered by the battery inverter: (Select all that apply)	☐ Oxygen Concentrator ☐ Oxygen Flow Splitter	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
		☐ CPAP ☐ Suction Pump ☐ Syringe Pump ☐ Phototherapy Light ☐ Radiant Warmer ☐ Incubator ☐ Other (<i>specify below</i>)	
			—
	Solar Power		
3.13	Does this facility have solar power?	☐ Yes ☐ No ☐ Yes, but not observed	If "No", 3.19
3.14	Is the neonatal unit covered by solar power?	☐ Yes ☐ No	—
3.15	Does the solar power exclusively cover the neonatal unit?	☐ Yes ☐ No ☐ Unknown	—
3.16	What neonatal unit lighting does the solar power cover?	☐ All lighting ☐ Some lighting ☐ No lighting	—
3.17	What neonatal unit equipment does the solar power cover?	☐ All equipment ☐ Some equipment ☐ No equipment	If All or No, 3.19
3.18	Specify what neonatal unit equipment is powered by the solar power: (<i>Select all that apply</i>)	☐ Oxygen Concentrator ☐ Oxygen Flow Splitter ☐ CPAP ☐ Suction Pump ☐ Syringe Pump ☐ Phototherapy Light ☐ Radiant Warmer ☐ Incubator ☐ Other (<i>specify below</i>)	—
			—
	Other		
3.19	Do you feel that power outages are impacting care on the neonatal unit?	☐ Yes ☐ No	—

Instructions: Direct these questions to the technician. For some questions, you will need to observe the staff or specific items/spaces at the facility. Please do not leave any variable blank. If a response is unknown, write "unknown" or select "unknown" from the multiple-choice answers.

No.	Item	Response	Skip to
	Outlets and Power Strips		
3.20	Given the amount of equipment on the unit, are electrical outlets sufficient? (<i>Sufficient means all equipment can be plugged in. Select No if devices or equipment ever need to be</i>)	☐ Yes ☐ No	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
	<i>unplugged from electrical outlets to accommodate other devices.)</i>		
3.21	How many outlets are in the neonatal unit? (Count each outlet in the neonatal unit. Count double outlets as two separate outlets.)	Number:	—
3.22	How many power strips/surge protectors are available for use in the neonatal unit?	Number:	If "0", 3.27
3.23	How many outlets are there among all of the power strips/surge protectors that are available for use ? (Please count all available spaces to connect devices)	Number:	—
3.24	How many power strips/surge protectors are in use on the neonatal unit?	Number:	—
3.25	How many devices are currently connected to any of the power strips/surge protectors? (Please count the number of devices connected to power strips/surge protectors at the time of the visit)	Number:	—
3.26	Which devices are connected to the surge protectors? (Select all that apply)	☐ Oxygen concentrator ☐ Oxygen flow splitter ☐ CPAP ☐ Suction pump ☐ Syringe pump ☐ Phototherapy light ☐ Radiant warmer ☐ Incubator ☐ Other (specify below)	—
			—
3.27	Do you have an inline voltage surge protector (e.g., AVS)?	☐ Yes ☐ No	—
3.28	For observation only: Are there cables running across the floor that can be tripped over?	☐ Yes ☐ No	—
Power Quality Assessment			
3.29	Is there a voltage stabiliser in the unit?	☐ Yes ☐ No ☐ Yes, but not observed	—
3.30	Was a power audit and certification for electric load done for the unit in the last 12 months?	☐ Yes ☐ No ☐ Unknown	—
3.31	Has any equipment broken in the last 12 months due to problems with the electricity?	☐ Yes ☐ No ☐ Unknown	—
3.32	Does the facility conduct internal power quality assessments?	☐ Yes ☐ No ☐ Unknown	If "No", 3.34
3.33	Does the facility experience power surges or lags? (Select all that apply)	☐ None ☐ Power surges ☐ Power lags	—

Facility Name:

Facility Code:

No.	Item	Response	Skip to
3.34	Given the equipment in the neonatal unit, is the amperage of the power supply appropriate?	☐ Yes ☐ No ☐ Unknown	—
3.35	<i>Make any comments regarding your observation on the amperage of the power supply.</i>		—

Comments

Facility Name:

Facility Code:

Section 4. Neonatal Unit Layout

Instructions: Please have all participants fill in their names and positions below, before completing the following forms in *Section 4. Neonatal Unit Layout*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 4. Neonatal Unit Layout

Instructions: Take at least 3 photos of the neonatal unit from different angles. Try to take photos when there are no caretakers in the room. Please respect the privacy and confidentiality of patients and their caretakers. When photographing the neonatal unit, avoid up-close photographs depicting the face of neonates, and ensure caretakers are not in the frame. Do not photograph the KMC ward if mothers are present. For each photo, label what areas (e.g. high-risk area, low risk area, area for those admitted from outside the facility, isolation room, KMC unit, waiting area) are pictured.

Please measure the dimensions of the neonatal unit in cm and draw a basic outline on paper. Include all rooms, including patient areas, the KMC unit, and the waiting area for caretakers. For the waiting area, include any seating such as chairs or benches. See the appendix on neonatal unit layout for instructions. If the neonatal unit is made up of multiple rooms, please draw each room on a separate sheet of paper. The first diagram should be an overview of all the rooms and subsequent drawings should be detailed diagrams of each room. For each diagram, label what areas are in the diagram (e.g. overall layout, detailed diagram of KMC unit).

Include the following items:

1. Room dimensions (in cm)
2. Doors
3. Windows
4. Sinks/water sources
5. Electrical plugs
6. Baby beds/cots
7. Beds intended for mothers
8. CPAPs
9. Oxygen concentrators/cylinders
10. Oxygen wall ports
11. Phototherapy lights
12. Suction pumps
13. Radiant warmers
14. Incubators
15. Hand sanitiser
16. Nurse's station
17. Sluice room
18. Indicate isolation areas

Facility Name:

Facility Code:

Section 5. Admission, Discharge, and Referral Criteria

Instructions: Please have all participants fill in their names and positions below, before completing the following forms in *Section 5. Admission, Discharge, and Referral Criteria*.

Respondents	
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position
Respondent Name (first and last name)	Respondent Position

Facility Name:

Facility Code:

Section 5. Admission, Discharge, and Referral Criteria

Instructions: Direct these questions to the nurse in charge. Either write a summary or take a photo of official guidelines if available.

Prompt: “Now I’m going to ask you about admissions, discharge, and referral criteria.”

What are the admission criteria for the neonatal unit?

What are the admission criteria for neonates admitted to the paediatric ward?
(Write N/A if neonates never admitted into paediatric ward)

What are the discharge criteria for the neonatal unit?

What are the criteria for referring a newborn from this facility to a higher-level facility?
(If you are already the highest level facility, please write N/A)

Facility Name:

Facility Code:

What are the admission criteria for the neonatal unit?

What are the criteria for referring a newborn from this facility to a lower level facility?

What are the admission criteria for the KMC unit?

What are the discharge criteria for the KMC unit?

Comments

Facility Name:

Facility Code:

Appendix A. Neonatal Unit Layout Guidance

Instructions: Draw the floorplan of the neonatal unit, including separate rooms (with a description of the purposes of each of the rooms), equipment, windows, doors & other infrastructural items (as shown in legend below). Draw the overall layout showing the relation of the rooms to each other, and each room on a separate page with the detailed items listed below.

Legend



door; draw on floor with arc following the way that door opens



door; draw on wall in door's location



window; draw to size and shape of windows. Draw all connected louvers as one window, but louvers separated by wood frames as separate windows



sockets/power sources with one plug



sockets/power sources with two plugs

B

infant bassinet/cot

KMC

KMC bed

S

sink

PT

phototherapy unit

O₂

oxygen concentrator

bC

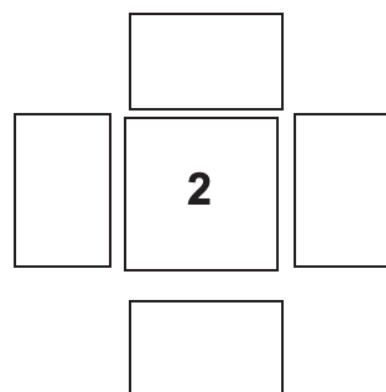
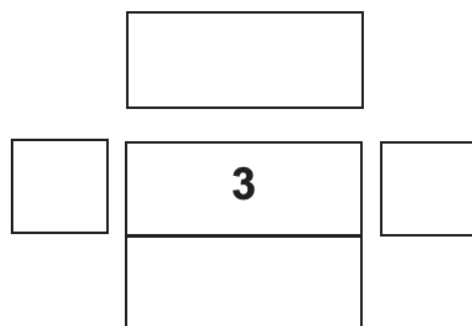
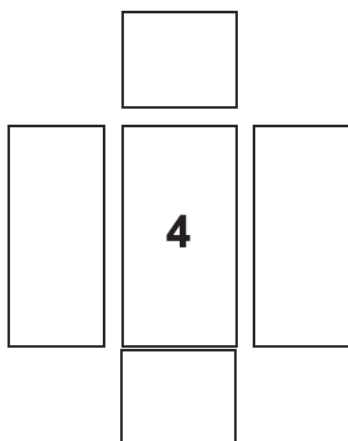
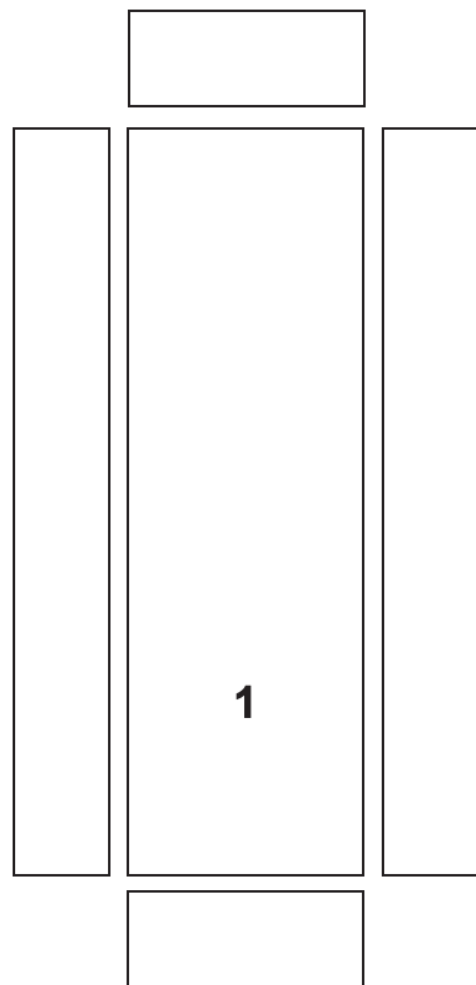
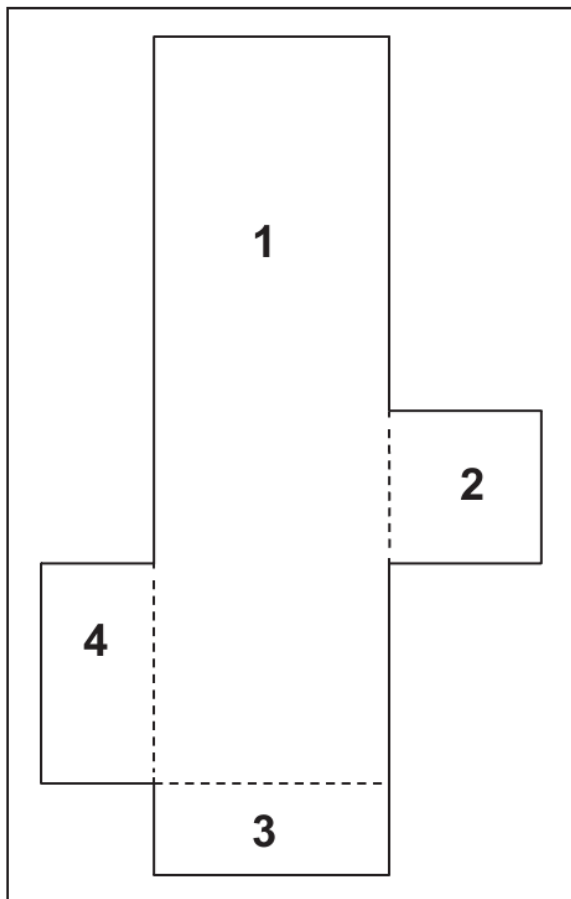
bCPAP machines

R

Radiant warmer

Facility Name:

Facility Code:



Facility Name:

Facility Code:

